

Failure Propagation in Multi-Agent LLM Systems: Characterization, Mechanisms, and Containment with Circuit Breakers

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Abstract

Large language models (LLMs) are increasingly deployed as compositional systems in which multiple agents interact to solve complex tasks. While such architectures enable modularity and improved capability, they introduce a new class of failure modes: errors originating in one component can propagate through the system, leading to cascade failures.

We study failure propagation in multi-agent LLM pipelines and evaluate circuit breakers as a mechanism for containment. We introduce Cascade Failure Rate (CFR) and provide both simulation and real-world validation.

Our results show that up to 62% of upstream failures propagate to final failure without protection, while circuit breakers can eliminate cascade failures entirely. We further show that adaptive containment strategies introduce a tradeoff between failure containment and task success.

We extend this analysis and show that this tradeoff is not inherent to circuit breaking, but is induced by degradation in fallback performance under load. When fallback quality remains high, adaptive strategies dominate; degradation introduces the observed tradeoff.

These findings suggest that reliability in compositional LLM systems is governed not only by component-level accuracy but by the dynamics of failure propagation and fallback behavior.

1 Introduction

Multi-agent LLM systems are increasingly used for planning, tool use, and reasoning. These systems improve capability but introduce new failure modes.

A key challenge is failure propagation: errors introduced in early stages can cascade through downstream components. Unlike isolated errors, these failures compound across stages.

Failure propagation scales multiplicatively with pipeline depth, making system reliability fundamentally different from component-level reliability.

We frame compositional LLM systems as distributed systems where reliability depends on controlling error propagation. Inspired by circuit breakers, we study mechanisms to contain failures before they cascade.

A central question is whether containment mechanisms introduce unavoidable tradeoffs. Prior observations suggest that reducing cascade failures comes at the cost of reduced task success. In this work, we revisit this assumption and show that the tradeoff emerges from fallback degradation rather than being inherent.

2 Theoretical Framework

Let p_u denote upstream failure probability and p_p denote propagation probability.

$$P(\text{final failure}) = p_u \cdot p_p \quad (1)$$

$$\text{CFR} = P(\text{final failure} \mid \text{upstream failure}) \quad (2)$$

For a pipeline of length L :

$$\text{CFR}(L) = 1 - (1 - p_p)^L \quad (3)$$

Even moderate propagation probabilities lead to high end-to-end failure rates in deeper pipelines. Circuit breakers reduce effective propagation by interrupting failure chains.

3 Method

We evaluate three configurations: no circuit breaker, AI-based circuit breaker, and adaptive circuit breaker.

We use a planner–executor–verifier pipeline. Failures are injected according to upstream probability p_u , and propagate with probability p_p .

An upstream failure is defined as invalid, incomplete, or inconsistent output.

We extend the model to include fallback behavior. When a breaker triggers, execution is routed to a fallback with success probability:

$$s_f = s_0(1 - \alpha \cdot \text{load}) \quad (4)$$

where α captures degradation strength.

4 Experiments

We evaluate pipelines across simulation and real-world settings.

Simulation sweeps propagation probability, degradation strength, and cost of catastrophic failure.

Real-world experiments use a planner–executor–verifier pipeline with structured outputs.

5 Results

5.1 Failure propagation

CFR increases with pipeline depth, consistent with theory.

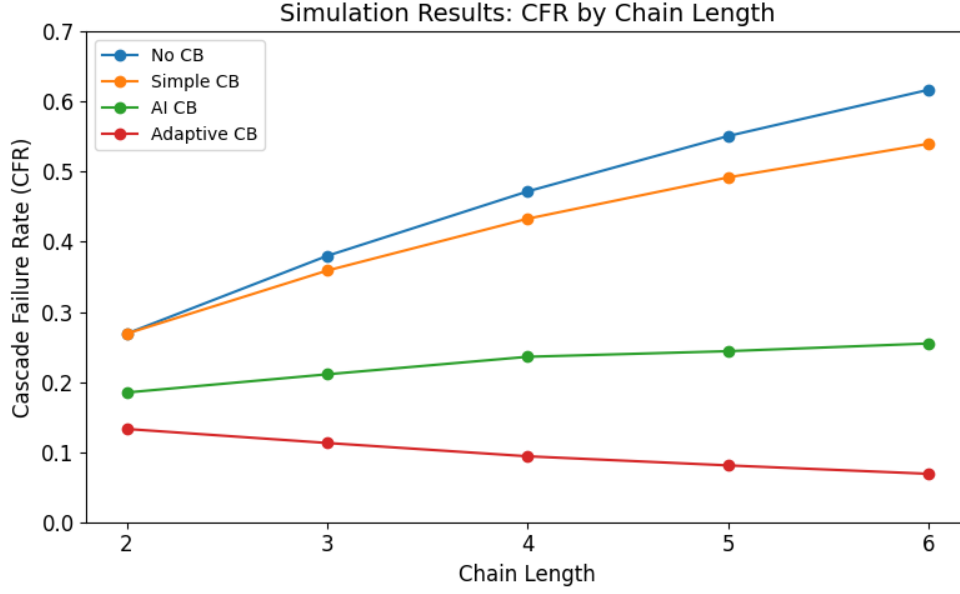


Figure 1: CFR increases with pipeline depth

Circuit breakers significantly reduce propagation.

5.2 Real-world validation

Method	Cascade Rate	Success
No CB	0.62	0.38
AI CB	0.00	0.44
Adaptive CB	0.00	0.40

Table 1: Real-world results

Without circuit breakers, 62% of upstream failures propagate. Both circuit breaker variants eliminate cascades.

5.3 Tradeoff mechanism

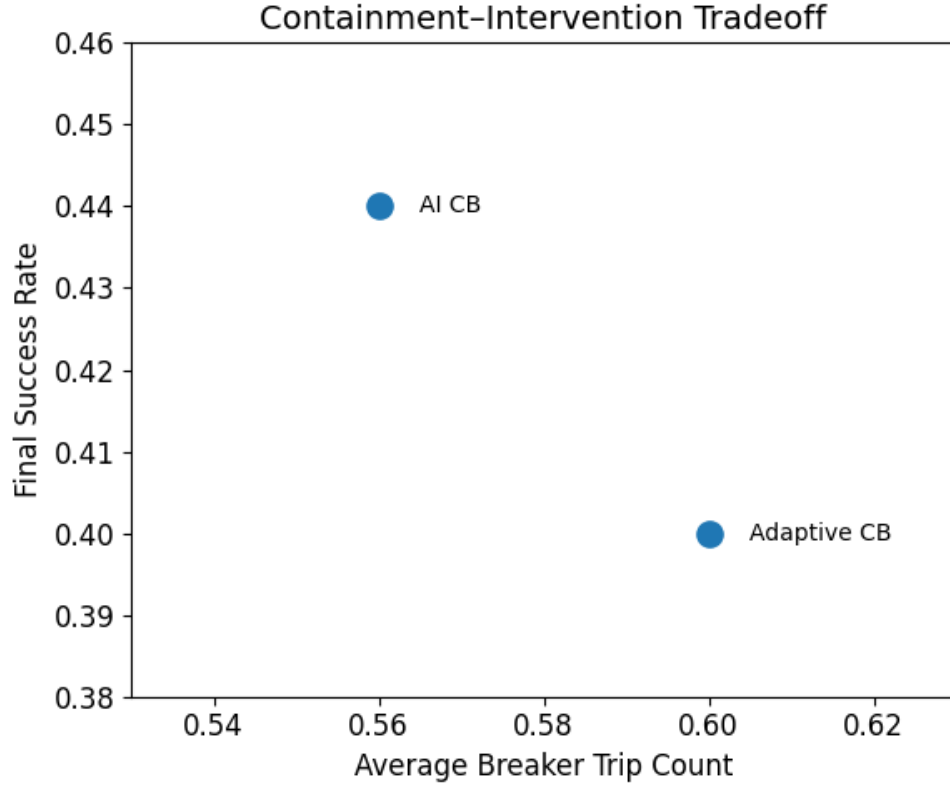


Figure 2: Containment vs success tradeoff

The tradeoff emerges when fallback quality degrades. With high-quality fallback, aggressive containment does not reduce success. As degradation increases, fallback failures reduce success.

5.4 Utility analysis

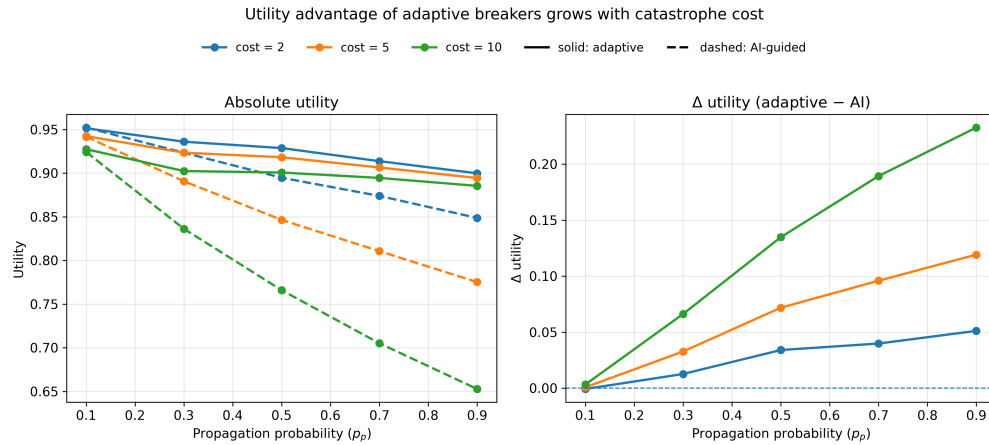


Figure 3: Utility increases with catastrophe cost

We define utility:

$$U = P(\text{success}) - c \cdot P(\text{catastrophic failure}) \quad (5)$$

Adaptive strategies achieve higher utility in high-risk regimes.

5.5 Robustness

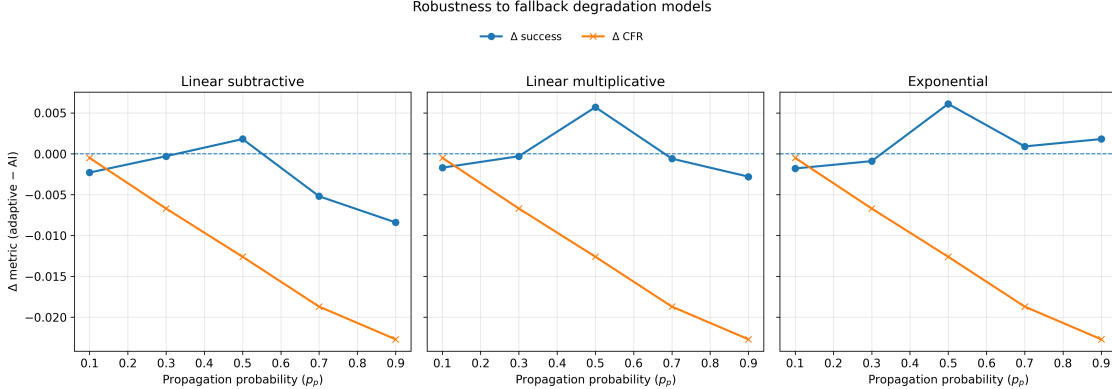


Figure 4: Robustness across degradation models

Results hold across degradation models.

5.6 Ablation

α	Δ Success	Δ CFR	Δ Utility
0.3	+0.002	-0.012	+0.034
0.5	-0.005	-0.018	+0.071
0.7	-0.012	-0.023	+0.119

Table 2: Effect of degradation strength

6 Discussion

Failure propagation is a system-level phenomenon. Reliability depends on controlling propagation and maintaining fallback quality.

7 Related Work

Prior work on multi-agent LLMs focuses on capability. Distributed systems introduce circuit breakers but assume binary failures. Our work models probabilistic propagation and degradation-induced tradeoffs.

8 Conclusion

Failure propagation governs reliability in LLM systems. Circuit breakers mitigate cascades, and adaptive policies improve utility. The safety–availability tradeoff is induced by fallback degradation.

References